

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Unit Test 1

Unit Test 1 - Answer Key

Class: Class 9

Assessment Type: Unit Test 1

Total Marks: 10

Duration: 30 minutes

Topics Covered: Python Fundamentals

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: MCQ (5 x 1 = 5 marks)

1. **Answer:** b) 512
 - **Explanation:** Exponentiation is right-associative. $2 ** 3 ** 2 = 2 ** 9 = 512$.
2. **Answer:** c) array
 - **Explanation:** array is not a built-in Python data type. Python has list, tuple, dict, and set.
3. **Answer:** a) el
 - **Explanation:** `x[:]` copies the string "Hello". `y[1] = "e"`, `y[3] = "l"`, so concatenation gives "el".
4. **Answer:** b) except
 - **Explanation:** Python uses try/except for exception handling. catch is used in other languages like Java/C#.
5. **Answer:** b) False
 - **Explanation:** `not True = False`. `False or False` and `True = False or False = False`. and has higher precedence than or.

Section II: Short Answer (5 x 1 = 5 marks)

1. **Write the output of the for loop.**
 - **Answer:** Output: 1 4 7
 - **Explanation:** `range(1, 10, 3)` generates 1, 4, 7 (stops before 10).
2. **Write a Python function `count_vowels(s)`.**
 - **Answer:**

```
def count_vowels(s):  
    vowels = "aeiouAEIOU"  
    count = 0  
    for ch in s:  
        if ch in vowels:  
            count += 1  
    return count  
  
print(count_vowels("Hello World")) # Output: 3
```
 - **Marking:** 1 mark for function definition, 1 mark for correct call and output.
3. **Difference between `range(5)` and `range(1, 5)`.**
 - **Answer:**
 - `range(5)` generates: 0, 1, 2, 3, 4 (starts from 0 by default)
 - `range(1, 5)` generates: 1, 2, 3, 4 (starts from 1, stops before 5)
4. **Write a program to check if a number is an Armstrong number.**
 - **Answer:**

```
def is_armstrong(n):
    digits = str(n)
    power = len(digits)
    total = sum(int(d) ** power for d in digits)
    return total == n

print(is_armstrong(153)) # True (13 + 53 + 33 = 1 + 125 + 27 = 153)
print(is_armstrong(123)) # False
```

5. What happens when you divide by zero in Python?

- **Answer:** Python raises a `ZeroDivisionError` exception. It can be handled using `try/except` blocks:

```
try:
    result = 10 / 0
except ZeroDivisionError:
    print("Cannot divide by zero!")
```

Viva Voce (5 marks)

1. Difference between a function definition and a function call. (1 mark)

- **Answer:** A function definition declares the function using `def` with a name and body. A function call executes the defined function by using its name followed by parentheses. E.g., `def greet():` is definition; `greet()` is call.

2. Explain mutable vs immutable data types. (1 mark)

- **Answer:** Mutable types (list, dict, set) can be modified after creation. Immutable types (int, float, string, tuple) cannot be modified; operations create new objects.

3. How does Python handle shallow copy vs deep copy of a nested list? (1 mark)

- **Answer:** A shallow copy creates a new outer list but references the same inner objects. A deep copy creates completely independent copies of all nested objects.

4. Purpose of `__init__` method in a Python class. (1 mark)

- **Answer:** `__init__` is a constructor method called automatically when an object is created. It initialises the object's attributes. Yes, an object can be created without it (uses default).

5. Difference between linear search and binary search. (1 mark)

- **Answer:** Linear search checks each element sequentially ($O(n)$). Binary search divides the sorted array in half each step ($O(\log n)$). Binary search requires a sorted list.